CONCEPT BRIEF

Intelligence-Driven Agentic Adversary Emulation

Putting an AI agent in the adversary's seat — grounded in real threat intelligence, bounded by hard controls

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Companion concept to the Agentic SecOps playbook · Discussion draft, June 2026 · Prepared as a thinking reference, not a build specification

The Idea in One Paragraph

Adversary emulation today is artisanal: a cyber threat intelligence (CTI) analyst reads an advisory on a named actor, hand-translates its tactics, techniques, and procedures (TTPs) into a test plan, and a red team builds the scenario — a process measured in weeks. The proposal here is to collapse that pipeline with an **agent that acts as the adversary**: it ingests the threat intelligence for a named actor (say, a CISA advisory on APT28), maps it to a structured MITRE ATT&CK technique set, and then executes that behavior in a sandboxed environment — emulating *that specific adversary*, not a generic attacker. Red emulation feeds blue detection; the loop reruns the day a new advisory drops. The contribution is not "AI that attacks" — it is making intelligence-driven emulation continuous, current, and auditable.

Why This Is Worth Pursuing

This concept sits at the convergence of three capabilities the playbook already treats as core: ATT&CK/ATLAS as the shared TTP grammar, autonomous adversary emulation, and the continuous purple loop. The value is concrete:

- **Speed and currency.** What takes a CTI-to-emulation team weeks becomes hours — and stays current, because the agent re-derives the emulation when intelligence changes.
- **Auditability.** Every action traces to a documented TTP from a named source, not red-team improvisation — the emulation is explainable to an auditor or Authorizing Official (AO).
- **Coverage at scale.** One analyst can run emulations for dozens of actors, re-baselining detection against the current threat landscape rather than a static annual exercise.
- **Purple-loop fit.** Continuous, intelligence-driven adversary emulation operationalizes the purple loop the playbook already advocates.

This Is an Emerging Discipline, Not a Blank Slate

A scan of the current landscape (mid-2026) shows the idea is real, named, and actively built — which is good news: the hard problems already have published solutions to borrow.

Agentic offensive frameworks already exist

- Academic systems RedTeamLLM and Incalmo demonstrate autonomous LLM-driven red teaming, Incalmo across multi-host networks; a hierarchical LLM–RL architecture decouples strategic planning (by a large language model, LLM) from tactical execution (by reinforcement learning, RL) along the ATT&CK kill chain.
- Commercial and training offerings have arrived: Dreadnode's agentic red-teaming system ("weeks to hours") and SANS SEC598 (AI red/blue/purple automation with adversary-emulation campaigns).

The CTI-to-emulation bridge is solved enough to use

- Extraction: IntelEX (retitled ThreatPilot), eLLM-CTI (STIX 2.1 output), and KGAgent4CTI reliably parse unstructured advisories into ATT&CK techniques — the exact ingestion step this concept needs.
- Emulation with fidelity control: Aurora generates attack chains guided by CTI reports, raising CTI-mentioned-technique coverage by ~30% over report-agnostic chains — and, critically, substitutes a close tool when one named in the report is unavailable while preserving the chain's logic.
- Evaluation: Microsoft's CTI-REALM benchmarks agents end-to-end against real attack telemetry with checkpoint scoring — a ready model for grading emulation fidelity rather than trusting it.

The Honest Risk Picture

Three failure modes separate a credible capability from a reckless one. Each has a mitigation the playbook is already built to supply.

Risk	Mitigation (already in the playbook's doctrine)
Dual-use / offense capability	An agent executing adversary TTPs is an autonomous offense capability. Mitigation: hard sandboxing, explicit authorization, and the playbook's rule that guardrails travel with the capability, not the intent. Emulation runs only in an isolated range, never production.
Fidelity drift (hallucinated TTPs)	An LLM will improvise beyond the documented behavior, corrupting the emulation's validity. Mitigation: CTI-grounded reward/constraint (the Aurora pattern) plus a separate evaluator agent grading each action against the source advisory — the adversarial-verification doctrine.
Certification of realism	How do you prove the agent emulated APT28 and not "generic attacker"? Mitigation: checkpoint-scored fidelity against the source TTP set (the CTI-REALM model), reported as a coverage metric alongside detection results.

A Reference Architecture (Conceptual)

Five stages, each with a deterministic boundary. This mirrors the SANS/Anthropic verification loop the playbook adopts for vulnerability analysis (discovery is cheap; a separate agent tries to disprove every finding), turned toward emulation:

1. **Ingest & map.** An extraction agent parses the threat-intel source (advisory, Mandiant/Volexity report) into a structured ATT&CK technique set with a confidence per technique. Output is reviewed before use.
2. **Plan.** A planner composes the techniques into a causally valid attack chain scoped to the target range — substituting available tooling where the named tool is absent, preserving chain logic.
3. **Authorize.** A human approves the plan and the blast-radius boundary. This is the hard gate — nothing executes without it.
4. **Execute (sandboxed).** An operator agent runs the chain only inside an isolated, instrumented range with egress controls — the agent-enclave pattern. Every action is logged.
5. **Evaluate & feed the loop.** A separate evaluator scores fidelity against the source TTP set; detections (or gaps) flow to blue as detection-as-code. Findings re-baseline the next run.

Deterministic boundaries do the safety work — the sandbox, the authorization gate, and egress control are non-LLM controls; the agent's judgment is never the thing keeping the operation contained.

Framing Is the Contribution

The right name is not "AI that attacks." It is **Intelligence-Driven Agentic Adversary Emulation** — TTP-grounded, sandbox-bound, fidelity-evaluated, human-authorized. That framing is what makes the capability defensible to leadership and an AO, and it is the through-line that connects this concept to the playbook's existing theses on dual-use, the offense/defense balance, containment-first design, and human-in-command operation. The technology is arriving regardless; the discipline around it is the part worth authoring. A proportionate first step: one advisory, one isolated range, one measured run, with fidelity and detection coverage reported to leadership before any expansion.

Sources are illustrative of the public state of the art as of mid-2026 and are listed inline. This brief is a discussion artifact for the Agentic SecOps playbook and is informational, not a build authorization or legal/operational guidance.

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